Financial Engineering '26

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Contents

1 Introduction **2**
1.1 Question at hand 2
1.2 Code structure 2

2 Analytical Approach **2**
2.1 Theoretical Background 2
2.2 Implementation 3

3 Monte Carlo **5**
3.1 Theoretical Background 5
3.2 Implementation 6

4 Vixification **6**
4.1 Theoretical Background 6
4.2 Implementation 7

5 Comparison of the Three Approaches **8**

1 Introduction

1.1 Question at hand

At maturity T, the buyer of a variance swap receives a payoff determined by the realised variance σ_R^2 accumulated over the life of the contract. Given a variance notional of N, the payoff is defined as:

$$\text{payoff}_T = N \cdot \left(\sigma_R^2 - K_{\text{var}}\right), \tag{1}$$

where K_{var} represents the variance strike (the fair delivery price such that the initial value of the contract is zero).

The realised variance σ_R^2, expressed in annual terms, is calculated using

$$\sigma_R^2 = \frac{252}{n} \sum_{i=1}^{n} \left(\ln\left(\frac{S_i}{S_{i-1}}\right)\right)^2. \tag{2}$$

Contract Details: We will assume that the maturity of the variance swap is 4 months. Furthermore, we set $S_0 = 100$, $r = 0.05$, and $q = 0$.

1.2 Code structure

The code, written in Python, is structured in a single class that encapsulates all logic for the different pricing approaches. The core of the `DerivativePricer` class is the `calibrate_model` method, which calibrates a pricing model using option price data (table 4). The calibrated model is then used in sections 3 and 4. This is achieved through several sub-methods (fig. 1). The `plot_performance` method prints and plots a variety of performance measures.

Finally, `monte_carlo` and `vixification` implement the logic for section 3 and section 4, respectively.

2 Analytical Approach

2.1 Theoretical Background

The Heston model specifies the following risk-neutral dynamics for the asset price S_t and its instantaneous variance v_t:

$$dS_t = \mu\, S_t\, dt + \sqrt{v_t}\, S_t\, dW_t^S, \tag{3}$$

$$dv_t = \kappa(\eta - v_t)\, dt + \theta\sqrt{v_t}\, dW_t^v, \tag{4}$$

where $\text{corr}(dW_t^S, dW_t^v) = \rho$. The five parameters are the mean reversion speed κ, the volatility of variance θ, the long-run variance η, the correlation ρ, and the initial variance v_0.

Under this model, the fair strike of a variance swap with maturity T admits the following closed-form approximation:

$$K_{\text{var}}^{AN} \approx \eta + (v_0 - \eta)\, \frac{1 - e^{-\kappa T}}{\kappa T} \tag{5}$$

Equation (5) has an intuitive interpretation: the fair strike is a maturity-weighted average of the initial variance v_0 and the long-run variance η. As $T \to \infty$, the factor $(1 - e^{-\kappa T})/(\kappa T) \to 0$ and $K_{\text{var}}^{\text{AN}} \to \eta$; for very short maturities the factor approaches 1 and $K_{\text{var}}^{\text{AN}} \to v_0$.

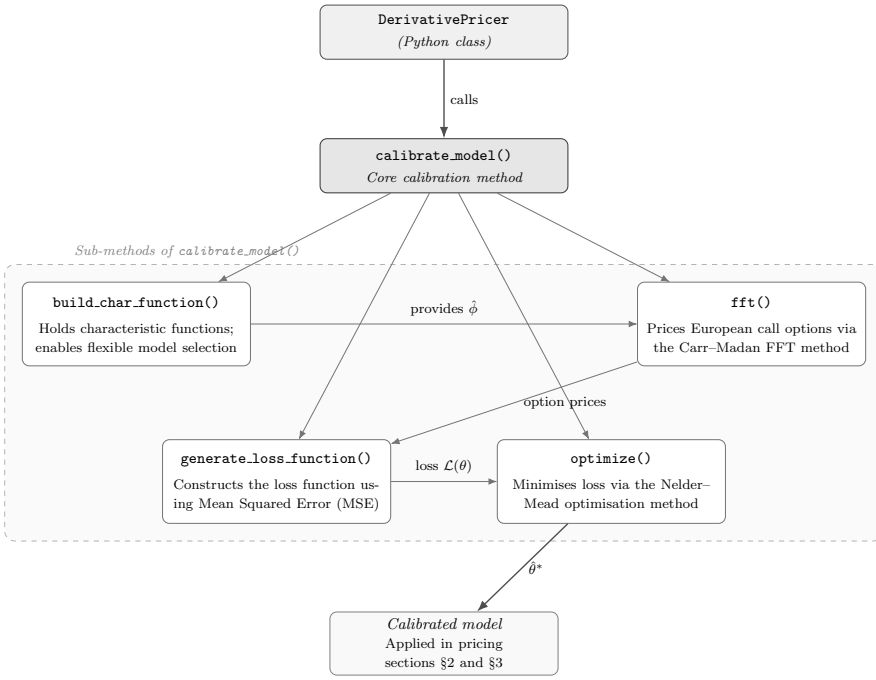

Figure 1: Schematic Overview `calibrate_model`

2.2 Implementation

Assumptions and choices

The model is calibrated on the assigned European call option data (table 4). Call prices are computed via the Carr–Madan formula, which evaluates the inverse Fourier transform of the Heston characteristic function using the FFT across the full strike grid. The loss function is the sum of squared errors (SSE) between model and market prices, minimised by the Nelder–Mead simplex algorithm. Given the five-dimensional parameter space, optimisation is restarted from seven distinct initial parameter vectors (table 1) to reduce the risk of converging to a local minimum. The FFT is applied on a grid of $N = 4096$ points, with frequency spacing $\eta = 0.25$ and damping parameter $\alpha = 1.5$. The choice N (a power of two) ensures efficient FFT execution. η provides sufficient resolution in log-strike space and α lies in the admissible range $\alpha > 1$ required for square-integrability of the modified call price.

Scenario	κ	θ	η	ρ	v_0
Baseline	1.0	0.3	0.04	-0.5	0.04
High mean reversion	2.0	0.5	0.06	-0.7	0.06
Low volatility regime	0.5	0.2	0.02	-0.3	0.02
High vol-of-vol	1.5	0.7	0.05	-0.6	0.05
High long-term variance	1.0	0.3	0.08	-0.5	0.08
Extreme correlation	1.0	0.3	0.04	-0.9	0.04
High initial variance	1.0	0.3	0.15	-0.5	0.15

Table 1: Starting values for Nelder–Mead calibration.

Parameter results

The results of the calibration are reported in table 2. The algorithm successfully converged after 26.79 seconds, yielding an RMSE of 0.0303 and a normalized RMSE of 0.0025. The latter indicates that, on average, the model deviates from observed option prices by approximately 0.25%. This is observable in fig. 2, where the lower graph plots the residuals. They are scattered relatively random, which implies low bias.

κ	θ	η	ρ	V_0
1.1211	0.2875	0.0623	-0.3772	0.0838

Table 2: Calibrated parameter values.

Feller condition

The Feller condition for the Heston model is given by

$$2\kappa\eta > \theta^2,$$

where κ denotes the mean reversion speed, η the long-run variance, and θ the volatility of variance.

Substituting the calibrated parameter values

$$2\kappa\eta \approx 0.1397 > 0.0827 \approx \theta^2,$$

the Feller condition is satisfied. Consequently, the variance process remains strictly positive almost surely in continuous time. However, in the Monte Carlo simulation of section 3, additional precautions are taken to ensure the positivity of the discretized variance process.

Fair Strike

Substituting the calibrated parameters into eq. (5) with $T = 4/12$ yields

$$K_{\text{var}}^{\text{AN}} = 0.0623 + (0.0838 - 0.0623) \cdot \frac{1 - e^{-1.1211 \times 1/3}}{1.1211 \times 1/3} \approx 0.0803.$$

Expressed as an annualised volatility, this corresponds to $\sqrt{0.0803} \approx 28.3\%$.

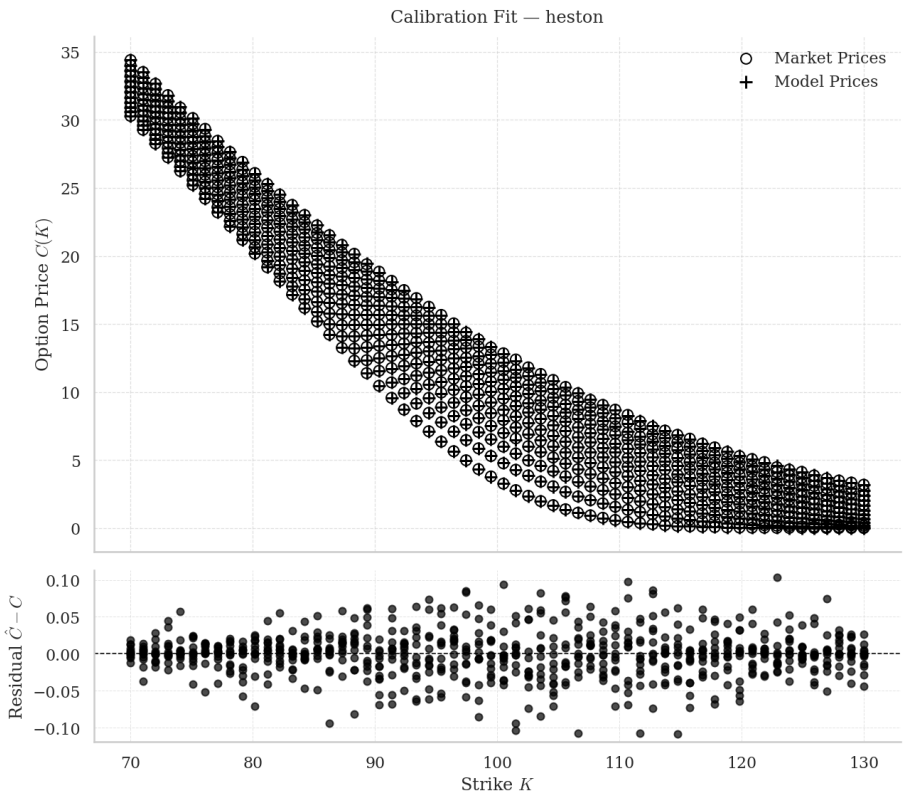

Figure 2: Heston model calibration diagnostics.

3 Monte Carlo

3.1 Theoretical Background

Under the risk-neutral measure $\mathbb{Q}$, the fair strike satisfies $K_{\text{var}}^{\text{MC}} = \mathbb{E}^{\mathbb{Q}}[\sigma_R^2]$. This follows directly from the zero-initial-value condition: the expected payoff $\mathbb{E}^{\mathbb{Q}}[\sigma_R^2 - K_{\text{var}}^{\text{MC}}] = 0$ implies $K_{\text{var}}^{\text{MC}} = \mathbb{E}^{\mathbb{Q}}[\sigma_R^2]$. This expectation is estimated by the sample mean

$$\widehat{K}_{\text{var}}^{\text{MC}} = \frac{1}{N} \sum_{j=1}^{N} \sigma_{R,j}^2, \tag{6}$$

where $\sigma_{R,j}^2$ is the annualised realised variance of the j-th simulated path. Since paths are independent, the estimator is unbiased and its standard error is $\widehat{\sigma}_R/\sqrt{N}$, yielding a 95% confidence interval of $\widehat{K}_{\text{var}}^{\text{MC}} \pm 1.96\,\widehat{\sigma}_R/\sqrt{N}$.

One limitation is discretisation bias: the Euler–Maruyama scheme accumulates squared log-returns over M discrete steps rather than integrating continuously, introducing a small downward bias in realised variance that vanishes as $M \to \infty$. A substantial improvement could be made by using the second order Milstein discretisation scheme.

3.2 Implementation

Assumptions and choices

Paths are simulated under the calibrated Heston dynamics using the Euler–Maruyama discretisation with $M = 84$ daily steps (one per trading day over the four-month horizon). Negative variance values arising from the discretisation are handled via reflection (`np.abs`), a simple positivity fix consistent with the full-truncation literature. The number of paths is set to $N = 10{,}000$, balancing precision against runtime.

Results

The method operates in two stages: `_simulate_paths` generates the Heston price paths, and `_calc_derivative_value` computes the realised variance for each path and returns its mean as the fair strike estimate. Applying this to the calibrated parameters from section 2 yields a fair strike of $\widehat{K}_{\text{var}}^{\text{MC}} = 0.0801$, or $\sqrt{0.0801} \approx 28.3\%$ in annualised volatility terms. The simulated paths are illustrated in fig. 3.

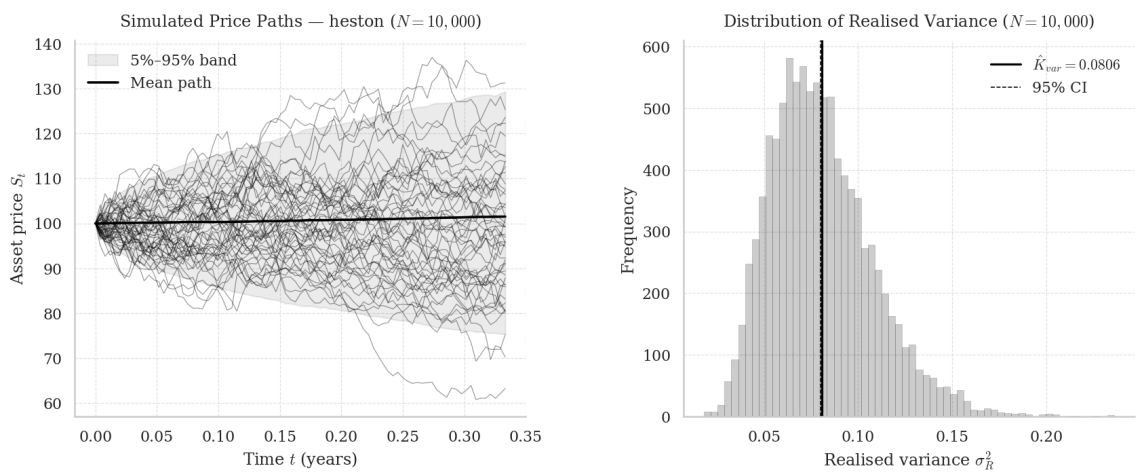

Figure 3: Simulated Monte Carlo paths

4 Vixification

4.1 Theoretical Background

Formally, the fair strike of a variance swap, $K_{\text{var}}^{\text{VX}}$, is given by (7):

$$K_{\text{var}}^{\text{VX}} = \frac{2e^{rT}}{T} \sum_i \frac{\Delta K_i}{K_i^2} Q(K_i, T) - \frac{1}{T} \left(\frac{F_T}{K_0} - 1 \right)^2 \tag{7}$$

Each term has a precise role. The summation is a discretised integral of out-of-the-money option prices, weighted by $1/K_i^2$, which reflects the fact that a log-contract can be spanned by an infinite

continuum of calls and puts. In practice, only a finite strip of strikes is available, so the integral is approximated by the midpoint quadrature rule. The correction term accounts for the gap between the forward price $F_T = S_0 e^{(r-q)T}$ and the nearest available strike below it, K_0; it vanishes when $K_0 = F_T$ exactly.

The function $Q(K_i, T)$ selects out-of-the-money option prices: put prices for strikes strictly below the forward, and call prices at or above it. This choice minimises sensitivity to bid-ask spreads, since OTM options are more liquidly quoted than ITM ones. Finally, the VIX itself is expressed as an annualised volatility in percentage points:

$$\text{VIX} = 100 \cdot \sqrt{K_{\text{var}}^{\text{VX}}} \tag{8}$$

4.2 Implementation

Assumptions and choices

Strike spacings ΔK_i are approximated via the midpoint quadrature rule. Only call prices are provided in the raw dataset (table 4). Put prices are recovered via put-call parity. A floor at zero is applied to $K_{\text{var}}^{\text{VX}}$ to guard against small numerical negatives arising from a sparse strike grid. The estimate is fully model-free: no calibrated parameters are required, and no assumption is made about the dynamics of the underlying.

Results

The forward price is $F_T = S_0 e^{(r-q)T}$. Strike spacings are computed via the midpoint rule:

$$\Delta K_i = \begin{cases} K_2 - K_1 & i = 1 \\ \dfrac{K_{i+1} - K_{i-1}}{2} & 1 < i < n \\ K_n - K_{n-1} & i = n \end{cases} \tag{9}$$

The OTM price selector assigns puts below the forward and calls at or above it:

$$Q(K_i, T) = \begin{cases} P(K_i, T) & K_i < F_T \\ C(K_i, T) & K_i \geq F_T \end{cases} \tag{10}$$

The put-call parity $C - P = S_0 e^{-qT} - K e^{-rT}$ conversion, is handled by the `put_call_parity` method. The relevant call prices are then passed directly to `vixification`:

```
mask = np.isclose(maturities, T)
K_var, vix = app.vixification(
    T,
    strikes      = strikes[mask],
    call_prices  = prices[mask]
)
```

Applying eq. (7) to the observed option cross-section at $T = 4/12$ yields $K_{\text{var}}^{\text{VX}} \approx 0.0787$, or $\sqrt{0.0787} \approx 28.1\%$ in annualised volatility terms. This estimate is slightly below the analytical value of 0.0803 and the Monte Carlo estimate of 0.0801. The small divergence is expected: the VIX

formula integrates over the available strike grid only, so truncation of the tails causes a modest underestimation of the true variance.

5 Comparison of the Three Approaches

The three approaches to pricing the variance swap differ along three axes: the assumptions they impose on the price process, the computational resources they require, and the structural sources of error they introduce. Table 3 summarises the results.

Approach	K_{var}	**Cost**	**Model-free?**
Analytical (section 2)	0.0803	<1 ms	No
Monte Carlo (section 3)	0.0801	∼2 s	No
VIX replication (section 4)	0.0787	<1 ms	**Yes**

Table 3: Comparison fair strike estimates at $T = 4/12$.

Analytical approximation. The closed-form formula (5) is the fastest and most transparent method. Its derivation exploits the affine structure of the Heston model and is exact for continuously monitored variance. The approximation arises from applying it to the discretely sampled realised variance of eq. (4), which accumulates squared log-returns over $n = 84$ steps rather than integrating v_t continuously.

Monte Carlo. The Monte Carlo estimate is unbiased in the limit $N \to \infty$ and $M \to \infty$, and it does not require the closed-form approximation step. Its disadvantages are twofold. First, it carries a sampling error of order $1/\sqrt{N}$, which for $N = 10{,}000$ is roughly ± 0.001 in variance units. Second, the Euler–Maruyama discretisation with $M = 84$ steps introduces a systematic downward bias in realised variance relative to the continuous-time path. The estimate is fully model-dependent, it inherits all parameter uncertainty from the Heston calibration.

VIX replication. The VIX replication formula is applied directly to observed market option prices, making it the only fully model-free estimate among the three. Its practical limitation is the truncation of the strike grid, if the available strikes do not extend far into the tails, the discretised integral underestimates the true variance.

Conclusion. For a quick, interpretable fair-value estimate, the analytical formula is preferred. Monte Carlo is valuable for pricing exotic path-dependent contracts where no closed form exists, and provides a direct simulation check. The VIX estimate, given a wide set of option data, is the most robust, as it is immune to model misspecification.

Table 4: Option Implied Volatility Surface (Maturities vs Strikes)

Maturities Strikes	0.083333	0.166667	0.250000	0.333333	0.416667	0.500000	0.583333	0.666667	0.750000	0.833333	0.916667	1.000000
70.000000	30.29	30.59	30.92	31.27	31.64	32.04	32.43	32.82	33.24	33.60	34.02	34.40
71.016949	29.28	29.58	29.92	30.29	30.70	31.09	31.52	31.90	32.36	32.73	33.14	33.52
72.033898	28.27	28.58	28.94	29.33	29.73	30.17	30.59	30.98	31.41	31.86	32.28	32.66
73.050847	27.25	27.58	27.95	28.36	28.78	29.21	29.66	30.13	30.50	30.93	31.38	31.82
74.067797	26.24	26.58	26.98	27.39	27.84	28.32	28.75	29.21	29.66	30.08	30.54	30.91
75.084746	25.23	25.58	25.99	26.42	26.91	27.36	27.87	28.37	28.76	29.23	29.69	30.14
76.101695	24.22	24.58	25.02	25.48	25.97	26.46	26.96	27.46	27.91	28.38	28.83	29.35
77.118644	23.21	23.59	24.06	24.54	25.06	25.57	26.08	26.57	27.09	27.59	28.02	28.47
78.135593	22.20	22.60	23.08	23.62	24.17	24.67	25.22	25.75	26.20	26.73	27.19	27.64
79.152542	21.19	21.62	22.14	22.69	23.24	23.79	24.38	24.89	25.41	25.92	26.38	26.92
80.169492	20.18	20.64	21.18	21.78	22.33	22.92	23.47	24.10	24.54	25.08	25.59	26.10
81.186441	19.17	19.69	20.25	20.86	21.44	22.10	22.64	23.25	23.73	24.27	24.80	25.28
82.203390	18.17	18.72	19.32	19.99	20.61	21.22	21.83	22.41	22.93	23.47	23.95	24.51
83.220339	17.17	17.77	18.43	19.07	19.77	20.39	21.00	21.61	22.16	22.73	23.27	23.75
84.237288	16.18	16.84	17.53	18.22	18.92	19.58	20.20	20.76	21.36	21.99	22.49	23.02
85.254237	15.19	15.91	16.65	17.37	18.06	18.76	19.40	20.03	20.60	21.24	21.69	22.25
86.271186	14.22	14.99	15.78	16.55	17.30	17.96	18.61	19.25	19.96	20.47	21.00	21.53
87.288136	13.25	14.09	14.91	15.74	16.52	17.14	17.86	18.50	19.11	19.70	20.27	20.82
88.305085	12.30	13.19	14.09	14.94	15.69	16.42	17.10	17.75	18.42	19.08	19.53	20.16
89.322034	11.39	12.34	13.32	14.17	14.96	15.68	16.38	17.00	17.64	18.26	18.92	19.39
90.338983	10.46	11.56	12.51	13.41	14.26	14.96	15.70	16.38	17.03	17.59	18.21	18.85
91.355932	9.57	10.70	11.76	12.64	13.45	14.20	14.98	15.62	16.33	17.01	17.54	18.18
92.372881	8.71	9.93	10.96	11.94	12.78	13.56	14.29	15.01	15.64	16.27	16.94	17.49
93.389831	7.87	9.15	10.30	11.19	12.10	12.91	13.70	14.35	15.04	15.63	16.19	16.90
94.406780	7.12	8.47	9.57	10.56	11.41	12.30	13.00	13.63	14.40	14.99	15.57	16.27
95.423729	6.35	7.78	8.89	9.96	10.73	11.59	12.38	13.02	13.78	14.36	15.00	15.67
96.440678	5.65	7.10	8.24	9.21	10.15	10.98	11.75	12.46	13.19	13.77	14.39	15.05
97.457627	4.98	6.47	7.66	8.66	9.63	10.44	11.16	11.88	12.52	13.26	13.76	14.37
98.474576	4.33	5.86	7.07	8.08	9.00	9.87	10.60	11.27	12.09	12.59	13.35	13.86
99.491525	3.79	5.29	6.53	7.52	8.44	9.25	10.07	10.77	11.49	12.07	12.77	13.28
100.508475	3.26	4.76	5.98	6.99	7.95	8.65	9.52	10.24	10.96	11.59	12.21	12.85
101.525424	2.79	4.29	5.48	6.59	7.40	8.22	9.01	9.67	10.43	11.05	11.64	12.39
102.542373	2.37	3.87	4.97	6.01	6.95	7.77	8.55	9.16	9.93	10.51	11.25	11.81
103.559322	1.98	3.42	4.57	5.61	6.40	7.18	8.04	8.82	9.36	10.10	10.67	11.33
104.576271	1.68	3.08	4.21	5.19	6.04	6.82	7.52	8.29	8.99	9.58	10.21	10.89
105.593220	1.38	2.67	3.81	4.74	5.52	6.33	7.12	7.85	8.43	9.07	9.80	10.38
106.610169	1.12	2.32	3.44	4.33	5.21	5.93	6.70	7.41	8.18	8.68	9.34	9.92
107.627119	0.93	2.04	3.07	3.99	4.81	5.52	6.33	6.99	7.66	8.26	8.88	9.43
108.644068	0.76	1.82	2.80	3.69	4.49	5.22	5.90	6.59	7.25	7.83	8.47	9.06
109.661017	0.59	1.57	2.50	3.30	4.15	4.84	5.58	6.19	6.94	7.47	8.02	8.70
110.677966	0.46	1.34	2.20	3.03	3.85	4.51	5.14	5.85	6.53	7.01	7.76	8.32
111.694915	0.37	1.19	2.00	2.80	3.46	4.22	4.86	5.57	6.17	6.73	7.43	7.85
112.711864	0.28	1.02	1.78	2.50	3.20	3.86	4.54	5.24	5.88	6.39	6.88	7.44
113.728814	0.22	0.88	1.60	2.33	2.97	3.58	4.30	4.89	5.52	6.04	6.62	7.17
114.745763	0.17	0.73	1.36	2.05	2.72	3.40	3.93	4.58	5.21	5.85	6.31	6.85
115.762712	0.12	0.62	1.24	1.88	2.47	3.20	3.76	4.32	4.87	5.48	5.99	6.57
116.779661	0.10	0.51	1.09	1.69	2.33	2.92	3.48	4.03	4.62	5.09	5.62	6.21
117.796610	0.07	0.45	0.94	1.54	2.10	2.74	3.23	3.74	4.39	4.82	5.36	5.93
118.813559	0.05	0.37	0.86	1.35	1.94	2.47	2.99	3.59	4.08	4.56	5.10	5.60
119.830508	0.04	0.31	0.74	1.20	1.79	2.27	2.85	3.38	3.88	4.34	4.81	5.29
120.847458	0.03	0.25	0.64	1.12	1.61	2.12	2.63	3.04	3.60	4.13	4.55	5.07
121.864407	0.02	0.21	0.55	0.98	1.46	1.92	2.44	2.90	3.39	3.87	4.32	4.85
122.881356	0.02	0.18	0.48	0.91	1.33	1.73	2.27	2.72	3.18	3.66	4.13	4.48
123.898305	0.01	0.15	0.44	0.79	1.18	1.60	2.07	2.51	3.02	3.46	3.88	4.30
124.915254	0.01	0.13	0.38	0.70	1.10	1.49	1.92	2.36	2.76	3.24	3.69	4.13
125.932203	0.01	0.10	0.32	0.63	1.01	1.40	1.82	2.16	2.62	3.09	3.48	3.87
126.949152	0.00	0.08	0.29	0.56	0.89	1.28	1.64	2.02	2.47	2.92	3.32	3.64
127.966102	0.00	0.07	0.24	0.50	0.79	1.17	1.49	1.89	2.28	2.70	3.14	3.49
128.983051	0.00	0.05	0.20	0.44	0.71	1.03	1.41	1.75	2.15	2.58	2.96	3.35
130.000000	0.00	0.05	0.17	0.40	0.68	0.97	1.30	1.70	2.03	2.37	2.76	3.19